

```

int led=3;

int trig =11;

int echo=10;

float myMap(float value, float fromLow, float fromHigh,
            float toLow, float toHigh)
{
    return (value - fromLow) * (toHigh - toLow)
        / (fromHigh - fromLow) + toLow;
}

void setup()
{
    pinMode(trig,OUTPUT);
    pinMode (echo,INPUT);
    pinMode(led, OUTPUT);
}

void loop()
{
    digitalWrite(trig,LOW);
    delayMicroseconds(2);

    digitalWrite(trig,HIGH);
    delayMicroseconds(10);
    digitalWrite(trig, LOW);

    int duration = pulseIn(echo,HIGH);
    int distance= duration*0.034/2;
    if (distance>=20 && distance <=200)

```

```

{
float input1= myMap (distance,20,200,0,255);
analogWrite(led,input1);
}
else
{
digitalWrite (led,LOW);
}
}

```

